PHYS1111 General Physics I Mid-term Exam
Date: 7 Nov 2026 (Sat)
Time: 4:00 – 5:30 PM (1hr 30 min)

- *Answer ALL 6 questions in Levels 1 and 2; the total score is 80 marks.*
- *Name the laws, theorems, or principles to be applied.*
- *Write down the formulae or equations (in standard symbols) to be applied.*
- *Show steps in calculations.*
- *Give final numerical values correct to 3 significant figures or 2 decimal places, in proper SI units.*
- *Take the free-fall acceleration g near Earth's surface as 9.81 m/s^2.*

Question 1 (10 marks)

Derive a general expression for the maximum height h and the range R of a ball kicked with an initial speed v_0 at an angle θ_0 above the horizontal (between 0 and 90°). We will assume that the ball is kicked from ground level and that it returns to ground level. In addition, for a given v_0, we want to calculate the value of θ_0 that gives the maximum height and the maximum horizontal range.

$1(a)$ *Max Height* $v_y = 0$

$$0^2 = \left(v_0 \sin\theta\right)^2 + 2(-g)H$$

$$v_0^{\,2}(\sin\theta)^2 = 2gH$$

$$H = \frac{v_0^{\,2}(\sin\theta)^2}{2g}$$

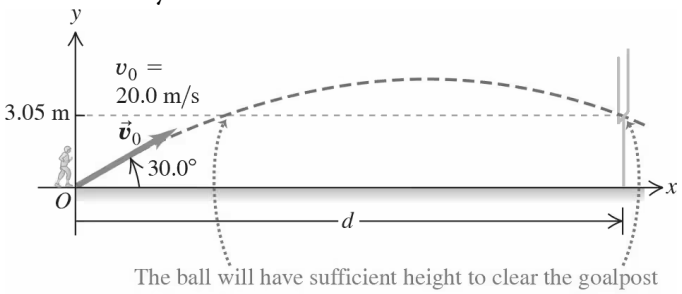
$1(b)$ *Max Range*

Horizontal $a_x = 0$	*Vertical* $x_y = 0$
$v_0 \cos\theta = \frac{R}{t}$	$0 = \left(v_0 \sin\theta\right)t + \frac{1}{2}(-g)t^2$
$t = \frac{R}{v_0 \cos\theta}$	$\frac{1}{2}gt^2 = v_0 \sin\theta\, t$
	$t = \frac{2v_0 \sin\theta}{g}$

$$\frac{R}{v_0 \cos\theta} = \frac{2v_0 \sin\theta}{g} \Rightarrow R = \frac{2v_0^{\,2}\sin\theta\cos\theta}{g}$$

$$\sin 2\theta = 2\sin\theta\cos\theta \Rightarrow R = \frac{v_0^{\,2}\sin 2\theta}{g}$$

We will look whether or not a projectile will clear a certain height at a specific point in its trajectory. Consider a field goal attempt, where a football is kicked from a point on the ground that is a horizontal distance d from the goalpost. For the attempt to be successful, the ball must clear the crossbar, 10 ft (about 3.05 m) above the ground, as shown below. The ball leaves the kicker's foot with an initial speed of 20.0m/s at an angle of 30.0° above the horizontal. What is the distance d between kicker and goalpost if the ball barely clears the crossbar on its way back down?



The ball will have sufficient height to clear the goalpost at two different values of x. However, only the height at $x = d$ describes the ball on its way back down.

$1(c)$ *Trajectory Equation*

$$v_0 = 20m/s,\ \theta = 30^\circ,\ y = 30.5m,\ d = ?$$

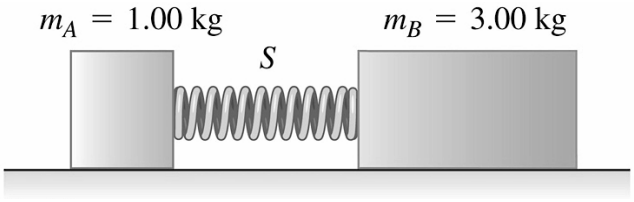
$$30.5 = x\tan(30) - \frac{9.81x^2}{2(20)^2(\cos 30)^2}$$

$$0.01635x^2 - \frac{\sqrt{3}}{3}x + 30.5 = 0$$

$$x = 6.46m(rej)\ or\ x = 28.8m$$

Question 2 (10 marks)

Block A (See Fig. 2) has mass 1 kg, and block B has mass 3 kg. The blocks are forced together, compressing a spring S between them; then the system is released from rest on a level, frictionless surface. The spring, which has negligible mass, is not fastened to either block and drops to the surface after it has expanded. Block B acquires a speed of 1.2 m/s. **(a)** What is the final speed of block A? **(b)** How much potential energy was stored in the compressed spring?



$2(a)\ v_{Af} = ?\ \ v_{Bf} = 1.2m/s$

$\rho_i = \rho_f$

$0 = (1)v_A + (3)(1.2)$

$v_{Af} = -3.6m/s$

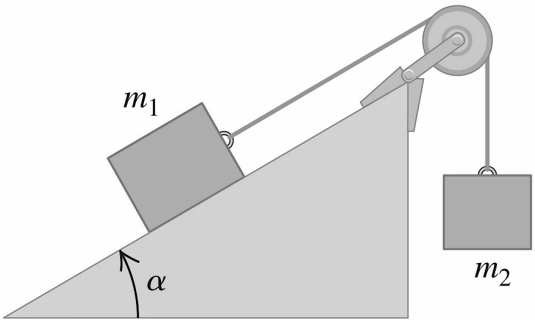
$2(b)\ EPE = ?\ EPE = KE$

$KE = \frac{1}{2}(3)(1.2)^2 + \frac{1}{2}(1)(3.6)^2$

$= 8.64J\ \ \therefore EPE = 8.64J$

Question 3 (20 marks)

A block with mass m_1 is placed on an inclined plane with slope angle α and is connected to a second hanging block with mass m_2 by a cord passing over a small, frictionless pulley (See Fig. 3). The coefficient of static friction is μ_s and the coefficient of kinetic friction is μ_k. **(a)** Find the mass m_2 for which block m_1 moves up the plane at constant speed once it is set in motion. **(b)** Find the mass m_2 for which block m_1 moves down the plane at constant speed once it is set in motion. **(c)** For what range of values of m_2 will the blocks remain at rest if they are released from rest?



$3(a)\ m_1 \gg m_2$

$Block\ 1\text{:}\ T - m_1\, g\, sin\alpha - \mu_k\, m_1\, g\, cos\alpha = 0$

$T = m_1\, g\, sin\alpha + \mu_k\, m_1\, g\, cos\alpha$

$Block\ 2\text{:}\ m_2\, g - T = 0$

$T = m_2\, g$

$\Rightarrow m_1\, g\, sin\alpha + \mu_k\, m_1\, g\, cos\alpha = m_2\, g$

$\Rightarrow m_2 = m_1\, sin\alpha + \mu_k\, m_1\, cos\alpha$

$3(b)\ m_1 \ll m_2$

$Block\ 1\text{:}\ m_1\, g\, sin\alpha - \mu_k\, m_1\, g\, cos\alpha - T = 0$

$T = m_1\, g\, sin\alpha - \mu_k\, m_1\, g\, cos\alpha$

$Block\ 2\text{:}\ T - m_2\, g = 0$

$T = m_2\, g$

$\Rightarrow m_1\, g\, sin\alpha - \mu_k\, m_1\, g\, cos\alpha = m_2\, g$

$\Rightarrow m_2 = m_1\, sin\alpha - \mu_k\, m_1\, cos\alpha$

$3(c)\ m_1\, sin\alpha - \mu_s\, m_1\, cos\alpha \leq m_2 \leq m_1\, sin\alpha + \mu_s\, m_1\, cos\alpha$

Question 4 (20 marks)

A roller coaster of 350 kg starts from rest at point A and slides down the frictionless loop-the-loop shown in Fig. 4. **(a)** How fast is this roller coaster moving at point B? **(b)** How large is the force exerted by the roller coaster on the track at point B?

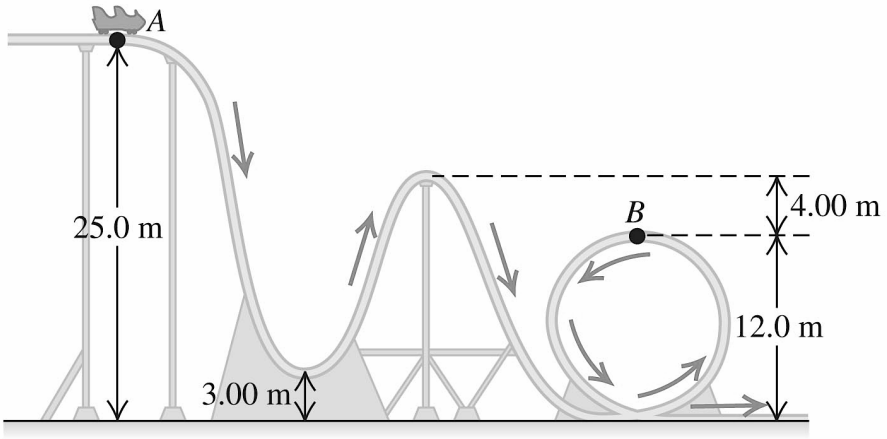


Fig. 4

$4(a)\ m_R = 350kg\ \ v_B = \ ?$

$W_A = \frac{1}{2}(350)(0)^2 + 350 \times 9.81 \times 25$

$W_A = 85800J$

$85800 = \frac{1}{2}(350)\ v_B^{\ 2} + 350 \times 9.81 \times 12$

$v_B = \pm 16.0\ m/s\ \ (take\ +\ 16m/s)$

$4(b)\ F_C = N + w$

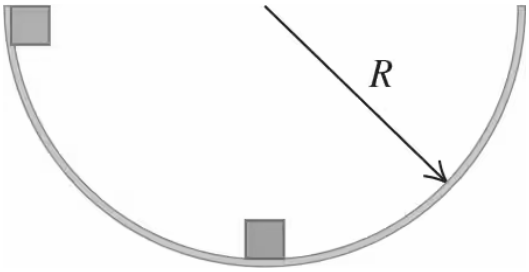
$\frac{350 \times 16^2}{6} = N + 350 \times 9.81$

$Normal = 1.15 \times 10^4 N$

Question 5 (10 marks)

A mass m is placed at the rim of a frictionless hemispherical bowl with a radius R and released from rest (Figure 8.49). It then slides down and undergoes a perfectly elastic collision with a second mass 3m sitting at rest at the bottom of the bowl.

a. What are the direction and speed of each mass just after the collision?
b. In terms of R, to what maximum height will each mass travel after the collision?



$5(a)\ Direction\ +\ Speed$

Cons of Energy

$mgR = \frac{1}{2}mv_{mi}^{\ 2}$

$v_{mi} = \sqrt{2gR}$

Velocity at bottom just before collision

Conservation of Momentum

$m\left(\sqrt{2gR}\right) + 3m(0) = m\,v_{mf} + 3m\,v_{3mf}$

$v_{mf} = \sqrt{2gR} - 3\,v_{3mf}$

Conservation of KE

$\frac{1}{2}m\left(\sqrt{2gR}\right)^2 + 0 = \frac{1}{2}m\,v_{mf}^{\ 2} + \frac{1}{2}(3m)\,v_{3mf}^{\ 2}$

$\frac{1}{2}m\left(\sqrt{2gR}\right)^2 + 0 = \frac{1}{2}m\left(\sqrt{2gR} - 3\,v_{3mf}\right)^2 + \frac{1}{2}(3m)\,v_{3mf}^{\ 2}$

$v_{3mf} = \frac{1}{2}\sqrt{2gR}$ *(to right)*

$v_{mf} = \sqrt{2gR} - 3\left(\frac{1}{2}\sqrt{2gR}\right)$

$v_{mf} = \ -\frac{1}{2}\sqrt{2gR}$ *(to left)*

$5(b)\ Max\ Height:\ KE_i = PE_f$

$m \Rightarrow \frac{1}{2}m\left(\frac{1}{2}\sqrt{2gR}\right)^2 = mgh$

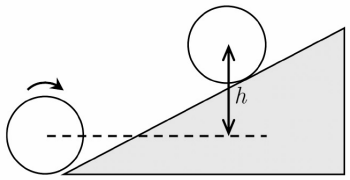
$h_m = \frac{1}{4}R$

$3m \Rightarrow \frac{1}{2}3m\left(\frac{1}{2}\sqrt{2gR}\right)^2 = 3mgh$

$h_{3m} = \frac{1}{4}R$

Question 6 (10 marks)

A 388-N wheel comes off a moving truck and rolls without slipping along a highway. At the bottom of a straight hill it is rotating at 23.4 rad/s (See the figure on the right). The radius of the wheel is 0.567 m, and its moment of inertia about its rotation axis is 0.8 MR^2. Friction does work on the wheel as it rolls up the hill to a stop, a height h above the bottom of the hill; this work has an absolute value of 3526 J. Calculate h.



$6: w = 388N \;\; \omega = 23.4 rad/s \;\; r = 0.567m \;\; I = 0.8MR^2 \;\; h = \; ?$

$m = \frac{W}{g} = \frac{388}{9.81} = 39.6kg$

$K_{rot} + K_{trans} - W_f = PE$

$K_{rot} = \frac{1}{2}\left(0.8 \times 39.6 \times 0.567^2\right)(23.4)^2 = 2788J$

$K_{trans} = \frac{1}{2}(39.6)(23.4 \times 0.567)^2 = 3485J$

$2788 + 3485 - 3526 = 39.6 \times 9.81 \times h$

$h = 7.07m$